

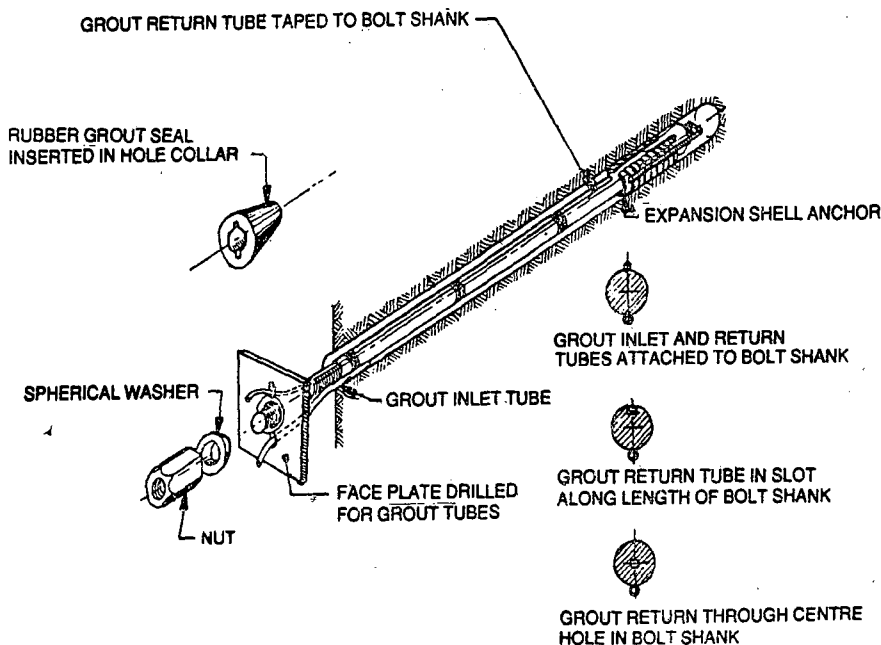
AMENDMENT NO. 2 OCTOBER 1992
TO
IS 5878 (Part 4) : 1971 CODE OF PRACTICE FOR
CONSTRUCTION OF TUNNELS CONVEYING
WATER

PART 4 TUNNEL SUPPORTS

(Page 19, clause 8.3.3) — Insert the following new clauses after 8.3.3:

8.3.4 Mechanically Anchored, Tensioned and Grouted Bolts

This generally consists of a wedge, attached to the bolt shank which is pulled into a conical anchor shell forcing it to expand against the drill hole walls. A rubber grout seal is used to centre the bolt in the hole and to seal the collar of the hole against grout leakage. Grout is then injected into the collar end of the hole and the return pipe is extended for the length of the hole. The grout injection is stopped when the air has been displaced and when grout starts flowing from the return tube (see Fig. 16). These type of bolts can be tensioned immediately after installation and grouted at a later stage when short term movements have ceased.



**FIG. 16 TYPICAL DETAILS OF MECHANICALLY ANCHORED,
TENSIONED AND GROUTED BOLTS**

Grouted, Tensioned Threaded Bar Bolts

This method of bolting consists of resin and catalyst contained in plastic capsules, the catalyst being separated in a glass or plastic container in the resin. These capsules are pushed into the hole with a loading stick and the bar is then inserted. Rotation of bar during insertion breaks the plastic containers and mixes the resin and catalyst (see Fig. 17). They are very convenient and simple to use. Very high strength anchors can be formed in rock of poor quality. These can be increasingly used in critical applications in which cost is less important than speed and reliability.

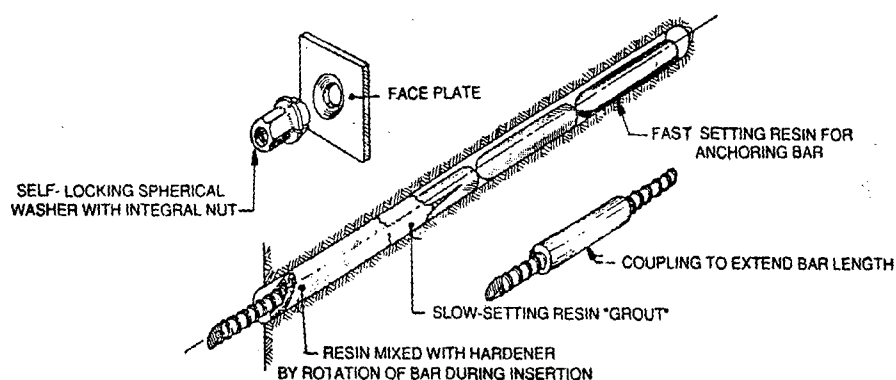


FIG. 17 TYPICAL DETAILS OF RESIN GROUTED, TENSIONED
THREADED BAR BOLTS

(RVD 14)

Printed at Printwell Printers, Aligarh, India